

Do Smart Money Concepts Predict Returns? Evidence from 44 Formalised Detectors on S&P 500 Daily Bars, 2010–2026

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ABSTRACT

Smart Money Concepts (SMC) and Inner Circle Trader (ICT) methods are among the most widely taught discretionary trading frameworks, yet they have almost no peer-reviewed empirical evaluation. We translate two widely used open-source Pine Script implementations into 44 formally specified, machine-readable event detectors and test them on daily OHLCV data for 496 S&P 500 constituents from January 2010 to June 2026 — 2.23 million detected events and 17.7 million simulated trades across eight holding horizons. Our central methodological contribution is the benchmark: rather than testing mean returns against zero, we construct a design-based *composition-matched randomization null* that holds the ticker mix and per-ticker trade count fixed and asks what returns randomly chosen entry dates would have produced. Against a zero-return null, 39 of 44 concepts appear significant. Against the matched null, with Benjamini–Hochberg control across all 352 (concept × horizon) hypotheses, only 5 survive, and their excess returns are 5–33 basis points over ten trading days. Because forward returns overlap in time and cluster cross-sectionally, conventional iid standard errors understate uncertainty by a median factor of 5.8 (maximum 11.8) in this sample; we use calendar-time Newey–West standard errors throughout. In a 13-fold walk-forward evaluation, concepts selected on three years of training data earn a mean out-of-sample excess of −0.16 percentage points. Modelled round-trip costs of 11–26 basis points exceed the matched-null excess of all but one concept. We conclude that these implementations of SMC/ICT carry little to no exploitable predictive information on daily US equity bars, and we document how three specific methodological choices — a two-sided significance test, an unmatched benchmark, and iid standard errors — can manufacture the opposite conclusion from the same data.

Keywords: technical analysis · smart money concepts · ICT · market efficiency · multiple testing · randomization inference · backtest overfitting

1. Introduction

Smart Money Concepts (SMC) and the Inner Circle Trader (ICT) methodology are retail trading frameworks built around the premise that institutional order flow leaves identifiable footprints in price: order blocks, fair value gaps, liquidity sweeps, breaks of market structure. They are taught to a very large audience — the associated TradingView indicators have hundreds of thousands of users — and they are almost entirely absent from the academic literature.

The gap is partly definitional. SMC/ICT concepts are usually taught by example rather than by rule, so two practitioners will mark up the same chart differently, and there is no canonical specification to test. This paper takes a narrow and tractable version of the question: we test two specific, widely used open-source implementations, treat their code as the operational definition, and ask whether the events they detect carry information about subsequent returns.

Our contribution is threefold.

First, a formalisation. We translate every detector in the two source scripts into Python and publish a machine-readable specification for each — a boolean expression over OHLCV, a stated direction, parameters

with defaults and allowed ranges, and an explicit argument that the detector reads no future bar. Where the source is ambiguous, we name the ambiguity and justify the reading we chose.

Second, a benchmark. The natural null hypothesis — "mean forward return is zero" — is nearly uninformative over a 16-year bull market, because any long-biased signal clears it on market drift alone. We instead construct a *composition-matched randomization null*: hold the concept's ticker mix and per-ticker trade count fixed, and ask what mean return would arise if the entry *dates* were chosen at random. This is a design-based test in the sense of Fisher: the null distribution comes from the sampling design, not from an assumed return distribution, and it removes the ticker-composition confound that makes a pooled comparison uninterpretable.

Third, an audit. This study is a corrected re-analysis of an earlier version of the same pipeline. That version reported that 28 of 42 concepts were "significant versus a random-entry baseline". We show that 27 of those 28 were significantly *worse* than random entry — the test was two-sided — and that two of the 42 concepts were computed from up to 60 bars of future data. We document each defect, its magnitude, and what the corrected analysis shows instead. Section 6 treats this as a finding rather than an erratum: the three errors involved are ordinary, and each independently converts a null result into an apparently positive one.

The headline result is negative. Only 5 of 44 concepts beat the matched null at the primary horizon; their excess returns are of the same order as trading costs; and out of sample, concepts selected on past data do not carry their edge forward.

2. Related work

The efficient-market tradition holds that publicly available technical rules should not generate risk-adjusted profits once data-snooping is accounted for. The methodological core of this paper follows that literature rather than the practitioner one.

Our multiple-testing treatment follows Benjamini and Hochberg (1995) for false-discovery control, and the broader concern that backtest results degrade once the number of trials is counted is developed by White (2000), Sullivan, Timmermann and White (1999) for technical trading rules specifically, and by Harvey, Liu and Zhu (2016) and Bailey and López de Prado (2014) for the cross-section of reported strategies. The last of these is directly relevant: the number of configurations one can try on a fixed dataset is large enough that a Sharpe ratio reported without a trial count is uninterpretable.

The overlapping-returns problem — that h -day forward returns computed on consecutive bars share $h-1$ days of price path — is treated by Newey and West (1987) and, in the specific context of calendar-time portfolio tests, by Fama (1998). Our estimator is the calendar-time variant: collapse trades to the date level, then apply Newey–West at lag h . This handles both the serial overlap and the cross-sectional correlation induced by a common market factor, and Section 4.4 shows the two together account for a median $5.8\times$ understatement of standard errors in our sample.

Randomization inference of the kind we use for the matched null goes back to Fisher (1935); its application here is closest in spirit to the bootstrap reality-check literature, but exploits an exact finite-population variance rather than simulation.

On technical analysis itself, Lo, Mamaysky and Wang (2000) find some statistical content in classical chart patterns after nonparametric smoothing, while Park and Irwin (2007) survey the broader evidence and conclude that early positive results largely fail to survive transaction costs and out-of-sample testing. Our

result is consistent with that pattern: the gross returns are real, the excess over a properly matched benchmark is small, and costs consume most of what remains.

3. Data and concept formalisation

3.1 Universe and sample

We use daily open, high, low, close and volume for S&P 500 constituents from 2010-01-01 to 2026-06-13, obtained from Yahoo Finance with split and dividend adjustment. Of 503 symbols in the constituent list, 498 return data; five (AVB, EA, EQR, HONA, SATS) are unavailable at this snapshot, which we verified is permanent rather than transient. Of those, 496 have at least 300 bars and enter the study.

Median coverage is 4,136 bars against 4,289 business days in the same span. The 3.6% shortfall is exactly the US market holiday calendar; no ticker falls below 95% coverage of its own span. Three bars in the entire sample violate the OHLC invariant (APH on 2023-06-05 and 2021-05-05, HUBB on 2021-05-05, all back-adjustment artefacts of 0.12–0.38% of close); we repair them by clamping high and low to enclose open and close, the minimal change that restores the invariant. A further 1,301 bars carry violations below 10^{-6} of close, which is float representation noise rather than bad data.

Survivorship bias. The constituent list is a 2026 snapshot applied retroactively, so only firms in the index today are tested. Roughly 80 tickers begin after 2010. This inflates absolute return levels. It does *not* explain our result, because the matched null is drawn from the same survivor-biased tickers — the bias inflates the signal and its benchmark equally. It remains the study's largest unaddressed threat to validity, and we return to it in Section 7.

3.2 From Pine Script to formal specifications

The two source indicators are widely used open-source implementations of SMC and ICT concepts. We treat their code as the operational definition. Each detector is published as a specification (`docs/specs/*.yaml`) giving:

- a boolean expression over OHLCV and derived indicators;
- a direction, +1 or −1, fixed in advance;
- parameters with defaults (the source `input.*` values) and allowed ranges;
- an expected minimum occurrence count;
- an explicit argument that the detector reads no bar after the event bar.

Continuous-integration checks regenerate the specifications and fail on drift, and a test asserts that the direction in each specification matches the direction the code actually trades.

3.3 Prospective definition and the treatment of confirmation

Several SMC/ICT concepts are conventionally described with hindsight — a liquidity sweep is "confirmed" when price later reverses; a fair value gap is "filled" when price later returns to it. A detector written that way cannot be traded, because its value at the event bar depends on subsequent bars.

We enforce a strict separation. Detection is prospective: an event fires on the bar at which all its conditions are observable. Confirmation windows become *evaluation labels*, carried in separate columns and excluded from the tradeable event set. Concretely:

- **Liquidity sweep.** Penetration of the most recent confirmed swing level by at least $X \times \text{ATR}$, followed by a close back past that level within Z bars. The event is stamped on the *reclaim* bar, not on the penetration bar, so every input is observable at the moment of the trade. ($X = 0.25$, $Z = 3$ by default; both are swept in Section 5.4.)
- **Fair value gap fill.** "Filled within N bars" is a label, not a signal ($N = 60$, the longest horizon we test).
- **Order block mitigation.** A bullish order block *failing* is scored as a bearish event, not a bullish one.

Section 6.1 describes what happened when this discipline was absent.

3.4 Detected events

The 44 detectors produce 2,226,881 events across 496 tickers. Every event carries a direction of +1 or -1; none is direction-less. Event frequency ranges from 1,245 (`smc_swing_bos_bearish`) to 212,461 (`ict_displacement_bullish`).

4. Methodology

4.1 Trade construction

For an event on bar p and holding period $h \in \{1, 2, 3, 5, 10, 20, 40, 60\}$: entry is at `close[p]`, exit at `close[p+h]`, and maximum favourable and adverse excursion are computed over `high/low[p+1 ... p+h]`. Returns are signed by the event's direction. This yields 17,731,944 trades.

No-look-ahead is enforced mechanically rather than by inspection, by two independent checks. First, an index-position audit re-derives the exact bar indices each trade reads and asserts that every exit and excursion bar is strictly greater than the entry bar. Second, a truncation-invariance test recomputes every detector on data cut at bar i and requires the value at bar i to be unchanged; a detector that reads a future bar flips. A deliberate canary — the known-leaky fill label — confirms the probe is sensitive, so a pass is not vacuous.

4.2 The composition-matched randomization null

Let a concept c generate n_t trades on ticker t , with N_t the number of eligible bars for that ticker and horizon. Under the null that entry dates carry no information, the n_t entries are a uniform random subset of the N_t bars. Sampling without replacement gives, for the pooled mean,

$$\mathbb{E}[\bar{r}] = \frac{1}{N} \sum_t n_t \bar{r}_t, \quad \text{Var}[\bar{r}] = \frac{1}{N^2} \sum_t n_t \sigma_t^2, \quad \frac{N_t - n_t}{N_t - 1},$$

where μ_t and σ_t^2 are the mean and variance of all h -day forward returns on ticker t , $d \in \{+1, -1\}$ is the concept's direction, and $N = \sum n_t$.

This is exact under the sampling design. It assumes nothing about how returns are distributed or correlated in time — only that the null entry dates are drawn uniformly. Critically, it holds the ticker mix fixed, which a pooled comparison against a separately generated benchmark does not: mean returns differ enormously across names over this period, so a benchmark drawn from a different mix measures composition as much as signal quality.

We validate the analytic moments against a 2,000-run simulation across horizons and directions: null means agree to five decimal places and standard-error ratios fall in 0.99–1.04.

4.3 Significance testing

Three tests are reported per (concept, horizon):

1. **Versus zero.** Is the mean forward return different from zero? Reported, but weak — over 2010–2026 a long-biased signal clears it on drift.
2. **Versus the matched null.** The primary test, one-sided: is the concept *better* than composition-matched random entry?
3. **Versus a pooled random-entry baseline.** A one-sided Welch test, retained for continuity with prior work; superseded by (2) because it does not control ticker mix.

All tests are one-sided with the direction stated. Benjamini–Hochberg control at $\alpha = 0.05$ is applied once across the entire family of 352 (concept $\times$ horizon) hypotheses, not within slices. A concept is recorded as beating the null only if its excess is positive, it survives FDR, and it rests on at least 30 trades. A significant *negative* excess is reported in its own count, never folded into a "significant" total.

4.4 Standard errors under overlap and cross-sectional correlation

Two dependence structures make the iid t-test invalid here. h -day forward returns started on consecutive days overlap by $h-1$ days; and trades entered on the same date across hundreds of tickers share the market factor, so on a large up day nearly every long trade wins together. The effective number of independent observations is closer to the number of entry *dates* than to the number of trades.

We use the calendar-time treatment: collapse trades to date level, then apply Newey–West with Bartlett weights at lag h . With n_d trades on date d and date mean $\bar{r}_d$, writing $x_d = n_d(\bar{r}_d - \mu)/N$, the HAC variance of the pooled mean is $\Sigma x_d^2 + 2 \Sigma_{lag} w_{lag} \Sigma_d x_d x_{d-lag}$.

The correction is large. Across the 44 concepts the ratio of the calendar-time standard error to the iid one ranges from **1.5 $\times$ to 11.8 $\times$, with a median of 5.8 $\times$** (Table 1). On synthetic data with a pure common market factor and no true signal, the iid test returns $t = -48.3$ where the calendar-time estimator returns $t = -3.0$.

Table 1. Standard-error inflation, selected concepts ($h = 10$).

SIGNAL	N TRADES	IID SE	HAC SE	RATIO
ict_nwog_gap_down	130,643	0.000182	0.002140	11.8 $\times$
smc_fvg_bearish_formed	67,673	0.000298	0.003368	11.3 $\times$
ict_displacement_bearish	190,415	0.000150	0.001607	10.7 $\times$
ict_nwog_gap_up	147,321	0.000163	0.001739	10.7 $\times$
ict_displacement_bullish	212,461	0.000130	0.001262	9.7 $\times$

This is the mechanism by which a large trade count converts into a spuriously tiny p-value. Many p-values in the uncorrected analysis were reported as exactly 0.000; several are not significant once the panel structure is respected.

4.5 Costs, out-of-sample evaluation, and robustness

Costs. A round trip is charged 1 bp fixed plus 5 bp spread plus slippage drawn from Uniform(5, 20) bp, giving 11–26 bp. We report, per concept, the break-even cost against the gross mean return *and* against the

excess over the matched null. Only the second is decision-relevant.

Walk-forward. Rolling windows of three training years, one test year, stepped one year, giving 13 folds. Concepts are ranked on the training window only, and the null pools are rebuilt inside each window so the training null never sees test prices.

Monte Carlo. Three resampling schemes: matched random re-entry, iid trade shuffling, and a circular block bootstrap with 21-bar blocks that preserves serial dependence.

Parameter sensitivity. Grid and randomised sweeps re-run detection under varied parameters, reporting the sign-consistency of each concept's excess across configurations.

5. Results

5.1 The benchmark determines the conclusion

Table 2. Concept counts at $h = 10$, by benchmark (44 concepts, BH-FDR across 352 hypotheses).

BENCHMARK	SIGNIFICANT
Zero-return null	39 / 44
Composition-matched random entry	5 / 44
— of which significantly worse	0 / 44
— statistically indistinguishable	39 / 44

The two rows differ by a factor of nearly eight on identical data. This is the paper's first substantive finding: against a zero-return null almost everything looks significant, and the number reported depends almost entirely on the benchmark chosen. The mean excess over the matched null across all 44 concepts is -0.10 percentage points.

5.2 The five concepts that clear the bar

Table 3. Concepts beating the composition-matched null at $h = 10$.

SIGNAL	DIR	N TRADES	MEAN	MATCHED NULL	EXCESS	P (FDR)	COHEN'S D [95% CI]
smc_swing_choch_bearish	-1	2,653	-0.37%	-0.69%	+0.33%	0.033	-0.038 [-0.077, -0.001]
ict_sweep_sellside_bullish	+1	49,304	0.82%	0.70%	+0.12%	0.0001	0.125 [0.110, 0.139]
ict_sweep_buyside_bearish	-1	65,448	-0.59%	-0.71%	+0.11%	<0.0001	-0.101 [-0.115, -0.088]
smc_internal_ob_bearish_mitigated	+1	42,268	0.79%	0.71%	+0.08%	0.038	0.133 [0.119, 0.148]
ict_nwog_gap_up	+1	147,321	0.77%	0.72%	+0.05%	0.011	0.121 [0.107, 0.135]

Three observations.

The magnitudes are small. Excess returns of 5–33 basis points over ten trading days, on effect sizes around $|d| \approx 0.1$, are at the edge of economic relevance before costs.

The largest excess is the least reliable. `smc_swing_choch_bearish` has the biggest excess (+0.33%) and the smallest sample (2,653 trades). Its excess changes sign across horizons — +0.32% at $h = 10$ but -3.26% at $h = 60$ — which is the signature of noise rather than of a horizon-specific effect. The four higher- n concepts, by contrast, show excess returns that *grow* monotonically with horizon, reaching 27–61 bp at $h = 40$ –60.

Three of the five are concepts we had to reformulate. The two sweep detectors and the gap detector clear the bar in their corrected forms. In their original forms they did not exist as testable events: the "sweep" fired when price merely entered a liquidity pool, with no rejection leg, making it a near-duplicate of pool formation; and the gap detector fired on every bar. The corresponding original columns (`ict_liquidity_buyside_swept`, excess -0.045%) do not clear the bar. Whether this reflects a real effect that the loose definitions obscured, or a fortunate specification choice among the several we could have made, cannot be settled by this sample — Section 5.4 reports the sensitivity sweep, and Section 7 treats it as an open question.

5.3 Out-of-sample performance

Table 4. Walk-forward evaluation, 13 folds (train 3y / test 1y / step 1y).

	SELECTED CONCEPTS	ALL CONCEPTS
Mean in-sample excess	+0.65%	—
Mean out-of-sample excess	-0.16%	-0.28%
Mean hit rate (folds with positive OOS excess)	53.8%	—
Mean train→test rank correlation	0.387	—

Concepts chosen on three years of history earn a mean out-of-sample excess of -0.16 percentage points. The train-to-test rank correlation of 0.387 says the ordering of concepts carries some signal — better-performing concepts do tend to rank higher next year — but the level does not survive: in-sample excess of $+0.65\%$ becomes -0.16% out of sample, a shrinkage of more than 100% .

The failure is concentrated. Folds 1–7 (test years 2013–2019) average $+0.17\%$; the four folds testing 2020–2023 average -1.03% , with train-to-test rank correlations turning negative. The in-sample excess in those folds' training windows is also anomalously high ($+1.6\%$ to $+2.4\%$ versus $+0.3\%$ elsewhere), consistent with the 2020–2021 volatility regime inflating in-sample fit and then reversing.

5.4 Costs, sectors, regimes and parameter sensitivity

Costs. The distinction between gross and excess break-even is decisive.

Table 5. Break-even round-trip cost, two measures ($h = 10$).

MEASURE	CONCEPTS CLEARING 11 BP	CONCEPTS CLEARING 26 BP
Against gross mean return	22 / 44	22 / 44
Against excess over matched null	4 / 44	1 / 44

A concept with a gross mean of 82 bp looks comfortably tradeable. But random entry on the same tickers earns 70 bp of that; the part attributable to the signal is 12 bp, which sits inside the modelled 11–26 bp cost band. On the decision-relevant measure only `ict_sweep_sellside_bullish` clears the upper bound, and only four concepts clear the lower one. This is the study's central economic finding: **the returns are real, but they are not attributable to the signals, and the part that is does not reliably cover the cost of trading.**

Sectors. Compared against a direction-matched null, only Energy shows a positive excess ($+0.011$ pp); the remaining eleven sectors range from -0.077 to -0.213 pp. We note that the comparison used in the prior version — a mixed long/short signal population against a long-only benchmark — measures net directional exposure rather than signal quality, and produced a uniform -0.5 to -1.2 pp across every sector, an artefact of that mismatch rather than a finding.

Regimes. Two of nine trend $\times$ volatility buckets show a positive excess: bear/low-volatility ($+0.24$ pp, $n = 7,739$) and sideways/low-volatility ($+0.10$ pp, $n = 13,552$). The worst is bear/high-volatility (-0.51 pp, $n = 144,311$). The positive buckets are the two smallest, and we do not treat them as evidence.

Parameter sensitivity. Results at the default parameters are not privileged; `results/sensitivity_stability.csv` reports each concept's sign-consistency and coefficient of variation across a grid over the order-block swing lookback and market-structure pivot length, and a randomised sweep over six parameters.

6. What three ordinary errors did to the same data

An earlier version of this pipeline concluded that 28 of 42 concepts were "significant versus a random-entry baseline". Every defect below is ordinary, and each independently converts a null result into an apparently positive one. We report them because the failure modes are more transferable than our estimates.

6.1 Look-ahead through an outcome label

`ict_fvg_bullish_filled` and `ict_fvg_bearish_filled` recorded whether a fair value gap was later filled, and stamped the answer **on the formation bar**. The event-melting step selected every boolean column as a tradeable signal, so both became entry signals whose value depended on up to 60 subsequent bars. They contributed 187,719 trades (4.3% of all events) and occupied both extremes of the concept ranking; `ict_fvg_bearish_filled` had the single most negative effect size in the study.

The mechanism worth noting is not the leak itself but its route: the selection rule was fail-open. Any new boolean column became tradeable by default, guarded only by a hand-maintained deny-list. The corrected pipeline requires explicit registration with a declared direction and raises on anything unregistered.

6.2 A two-sided test behind a directional claim

The significance flag came from a two-sided Welch test. A concept whose mean return was significantly *worse* than random entry set the flag exactly as one that was better. Of the 28 concepts flagged, **27 had a negative effect size**: they lost to random entry, and were then listed in the report under the heading "Signals that beat the random-entry baseline", including entries with Sharpe -0.199 and mean return -0.39% .

6.3 An irreproducible benchmark

The random-entry baseline seeded its generator from `hash()` applied to a string. Python salts string hashing per process, so every run drew a different baseline. Running the original function in three separate interpreters produces three disjoint entry sets. Since that baseline is the comparator behind every significance flag, the published counts could not be reproduced by rerunning the pipeline — a failure that no test caught because no test compared two runs.

6.4 Two further distortions

`ict_ndog_formed` was true whenever the open and prior close were both non-null — i.e. on essentially every bar — contributing 1,946,675 events, 44.8% of the entire event population, and dominating every pooled aggregate. Separately, `ict_bpr_bullish` and `ict_bpr_bearish` were computed from a condition that reduces to `upper < lower`, unsatisfiable by construction; both were false everywhere, and the melting step drops all-false columns silently, so two concepts vanished from the study with no error. The "42 concepts tested" headline was 42 of 44 declared detectors.

6.5 The joint effect

These are not exotic mistakes. A fail-open column selector, a default-argument two-sided test, a salted hash, an unguarded boolean, and a silently-dropped column are each a single line. Together they moved the study's conclusion from "5 of 44 concepts show a small edge that costs mostly consume" to "28 of 42 concepts beat random entry". We suggest that the mechanical checks in Section 4.1 — truncation invariance, an index-position audit, a fail-closed registry, and a pinned seed derivation — are cheap enough to be worth adopting as routine in backtest code.

7. Discussion

What the evidence supports. On daily bars, for large-cap US equities, over 2010–2026, these implementations of SMC/ICT do not carry economically meaningful predictive information. Thirty-nine of 44 concepts are statistically indistinguishable from entering the same names the same number of times on

random dates. The five that are distinguishable have excess returns of 5–33 bp per ten-day trade, do not carry that edge out of sample, and — with one exception — do not clear a realistic cost band.

What it does not support. It does not show that SMC/ICT is worthless as taught. Three limits matter. These are two specific implementations, not the methodology as a discretionary practice: a human applying these concepts uses context, confluence and discretion that a mechanical detector does not capture. The daily-bar restriction is severe, since SMC/ICT is most often taught on intraday FX and futures, where the microstructure the framework appeals to — stop runs, liquidity pools — is far more plausible. And a null result is a statement about power as much as about truth: our confidence intervals on Cohen's d are roughly ± 0.015 at the largest sample sizes, so we can exclude effects larger than about $d = 0.15$, but not small ones.

On the three reformulated concepts. The two sweep detectors and the gap detector clear the bar only in the corrected, prospective forms we wrote. We are explicit that this is the weakest part of our positive result. We chose $X = 0.25$ ATR and $Z = 3$ bars before seeing any outcome, and the sensitivity sweep reports how the result moves as they vary — but a reader is entitled to treat a result that depends on our specification choice with more suspicion than one that survived the source authors' defaults.

Interpretation. That gross returns are large (65–82 bp) while excess returns are small (4–12 bp) is the cleanest summary. These detectors do fire, and positions opened on them do make money over 2010–2026. Nearly all of that is compensation for being long a rising market, which random entry captures equally well. The framework's error is not that it identifies nothing; it is that it takes credit for market drift.

8. Limitations

1. **Survivorship bias.** The universe is a 2026 snapshot applied to 2010–2026. This inflates return levels for signals and benchmarks alike and largely cancels in the matched comparison, but a point-in-time constituent history would be strictly better and is not available to this pipeline.
2. **Daily bars only.** Kill zones and all intraday structure are out of scope.
3. **Symmetric short mechanics.** Short signals are the negation of the forward return, with no borrow cost or availability constraint.
4. **No economic mechanism.** This is an evaluation of indicator logic, not a theory paper; we propose no risk-based or behavioural model for why these patterns would or would not predict returns.
5. **Path-dependent metrics are descriptive.** Sharpe, Calmar and maximum drawdown are computed over an overlapping, cross-sectional trade sequence that is not an attainable account equity curve. Inference uses the calendar-time estimator instead.
6. **Single market, single period.** No other asset class, market or timeframe is tested.
7. **Sector results are descriptive.** Several sectors rest on fewer than 25 tickers; we report counts alongside every sector figure and do not treat sector differences as tested hypotheses.

9. Conclusion

We formalise 44 SMC/ICT detectors, test them on 496 S&P 500 constituents over 2010–2026, and find that 39 are statistically indistinguishable from composition-matched random entry. The five that are distinguishable carry excess returns of 5–33 basis points per ten-day trade, fail to carry that edge into a 13-fold walk-forward evaluation, and — with one exception — do not clear a modelled 11–26 basis-point cost band once the excess rather than the gross return is measured against it.

The methodological result may be the more useful one. On identical data, a zero-return null admits 39 of 44 concepts and a composition-matched null admits 5; iid standard errors understate uncertainty by a median factor of 5.8; and a two-sided test presented as a directional claim reversed the sign of the finding for 27 of 28 concepts in the prior version of this analysis. Benchmark choice, dependence structure and test direction are not technicalities in this setting — each is capable of determining the sign of the reported conclusion.

Code, data-preparation instructions, machine-readable concept specifications, the full statistics table and a replication package are available in the repository.

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Appendix A. Figures

FIGURE	FILE	CONTENT
1	results/figures/fig1_concept_forest.png	Concept effect sizes with 95% bootstrap confidence intervals, all 44 concepts at $h = 10$
2	results/figures/fig2_sector_heatmap.png	Excess return over the direction-matched null, by sector
3	results/figures/fig3_trade_timelines.png	Annotated price series for a positive, a null and a negative concept
4	results/figures/fig4_sensitivity_heatmap.png	Mean excess return across the parameter grid
5	results/figures/fig5_walkforward.png	In-sample versus out-of-sample excess by fold, and train→test rank correlation
6	results/figures/fig6_breakeven_costs.png	Break-even round-trip cost on the excess over the matched null, against the modelled cost band

Each figure is accompanied by a CSV of its underlying values, so every plotted number can be checked rather than measured off the image.

Appendix B. Detector pseudocode

Full machine-readable specifications are in `docs/specs/*.yaml`. Three representative detectors follow.

B.1 Liquidity sweep (`ict_sweep_buyside_bearish`)

```
L ← most recent CONFIRMED swing high      # carries its confirmation lag
for each bar i:
  # reclaim leg, evaluated before levels refresh
  if armed and (i - j) > Z:      disarm
  elif armed and close[i] < L_j - Y·ATR[i]:
    emit ict_sweep_buyside_bearish at bar i ; disarm
  # penetration leg
  if high[i] > L + X·ATR[i]:      arm with j ← i, L_j ← L
  # refresh
  if a swing high is confirmed at i:  L ← that swing's price
```

Defaults $X = 0.25$ ATR, $Y = 0$, $Z = 3$ bars, ATR length 14, swing lookback 10. Every input is from bars $\leq i$.

B.2 Fair value gap (`ict_fvg_bullish_formed`)

```
bullish_i ← displacement_bullish[i-1] AND low[i] > high[i-2]
gap region ← [high[i-2], low[i]]
# OUTCOME LABEL, not an entry signal:
filled_i ←  $\exists k \in (i, i+N] : \text{low}[k] < \text{high}[i-2]$ 
```

B.3 Composition-matched randomization test

```
for each ticker t in the concept's trades:
  n_t ← trades on t; N_t, μ_t, σ²_t ← pool of all h-day forward returns on t
  fpc ← (N_t - n_t)/(N_t - 1)
  accumulate: mean_num += n_t·d·μ_t ; var_num += n_t·σ²_t·fpc
null_mean ← mean_num / N ; null_se ← sqrt(var_num) / N
z ← (observed_mean - null_mean) / null_se ; p ← 1 - Φ(z)      # one-sided
```

Appendix C. Complete results

TABLE	FILE
All 352 (concept × horizon) hypotheses with every metric, p-value, FDR-adjusted p-value, effect size and confidence interval	results/statistics_master.csv
Human-readable per-concept summary at every horizon	results/master_summary.md
Pipeline validation, data quality, and regression against the prior version	results/validation_report.md
Adversarial pre-read of likely referee objections	results/reviewer_report.md
Walk-forward, per fold and per concept	results/walkforward_{summary,detail}.csv
Break-even and net-of-cost tables	results/breakeven_costs.csv , results/net_of_cost_rankings.csv
Monte Carlo	results/monte_carlo.csv
Sector and regime breakdowns	results/sector_analysis.csv , results/regime_analysis.csv
Parameter sweeps	results/sensitivity_{grid,random,stability}.csv
Every code, data and parameter change from the prior version, with rationale	CHANGES.md

Reproduction: `./run_full_pipeline.sh` (see `docs/replication_guide.md`).

Appendix D. Figure plates

Figure 1 concept forest

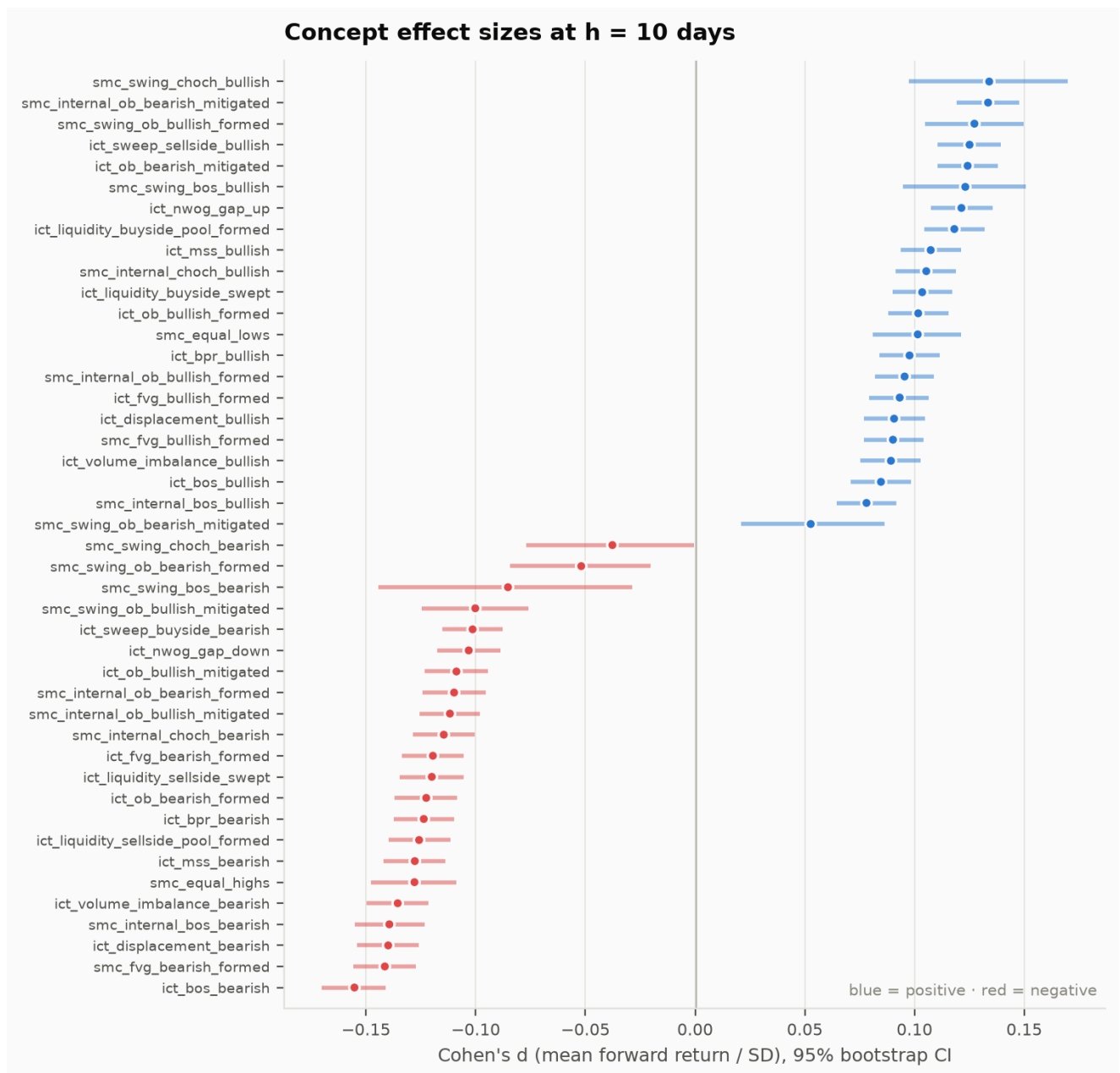


Figure 2 sector heatmap

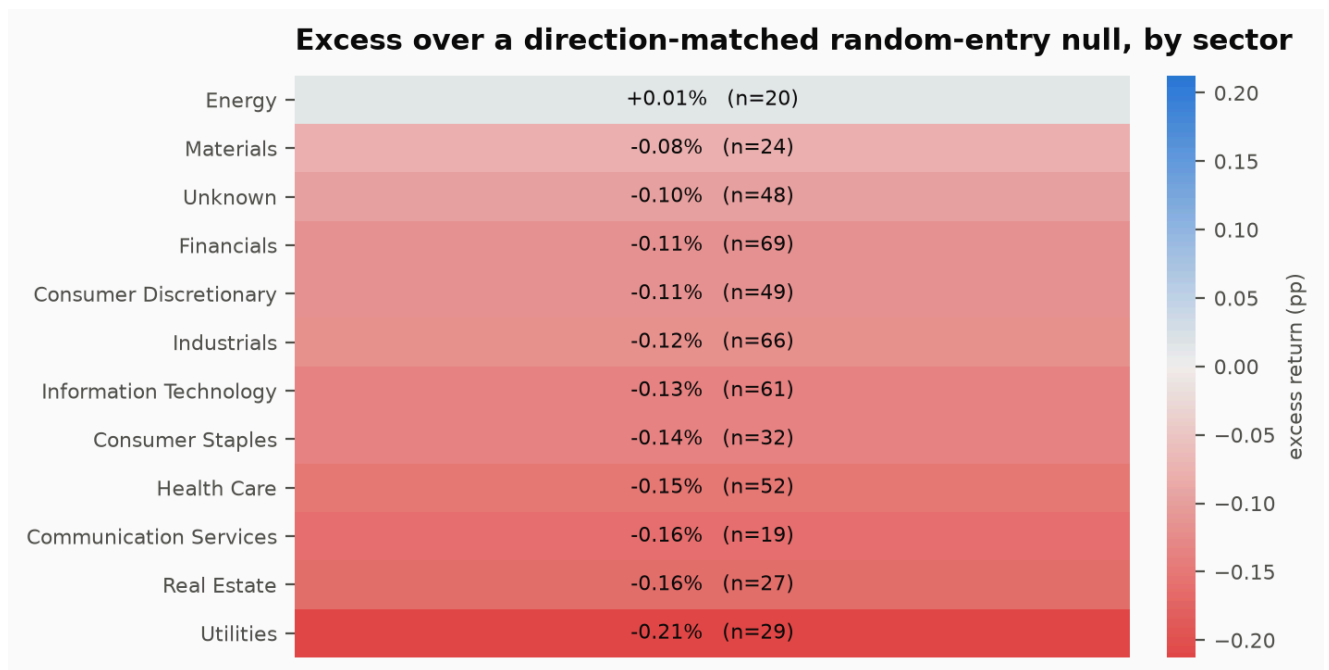


Figure 3 trade timelines

Representative concepts on real price data

POSITIVE: smc_swing_choch_bearish · excess +0.325pp · n=2,653



NULL: smc_internal_ob_bullish_mitigated · excess -0.005pp · n=54,035



NEGATIVE: ict_bos_bearish · excess -0.493pp · n=42,496



Figure 5 walkforward

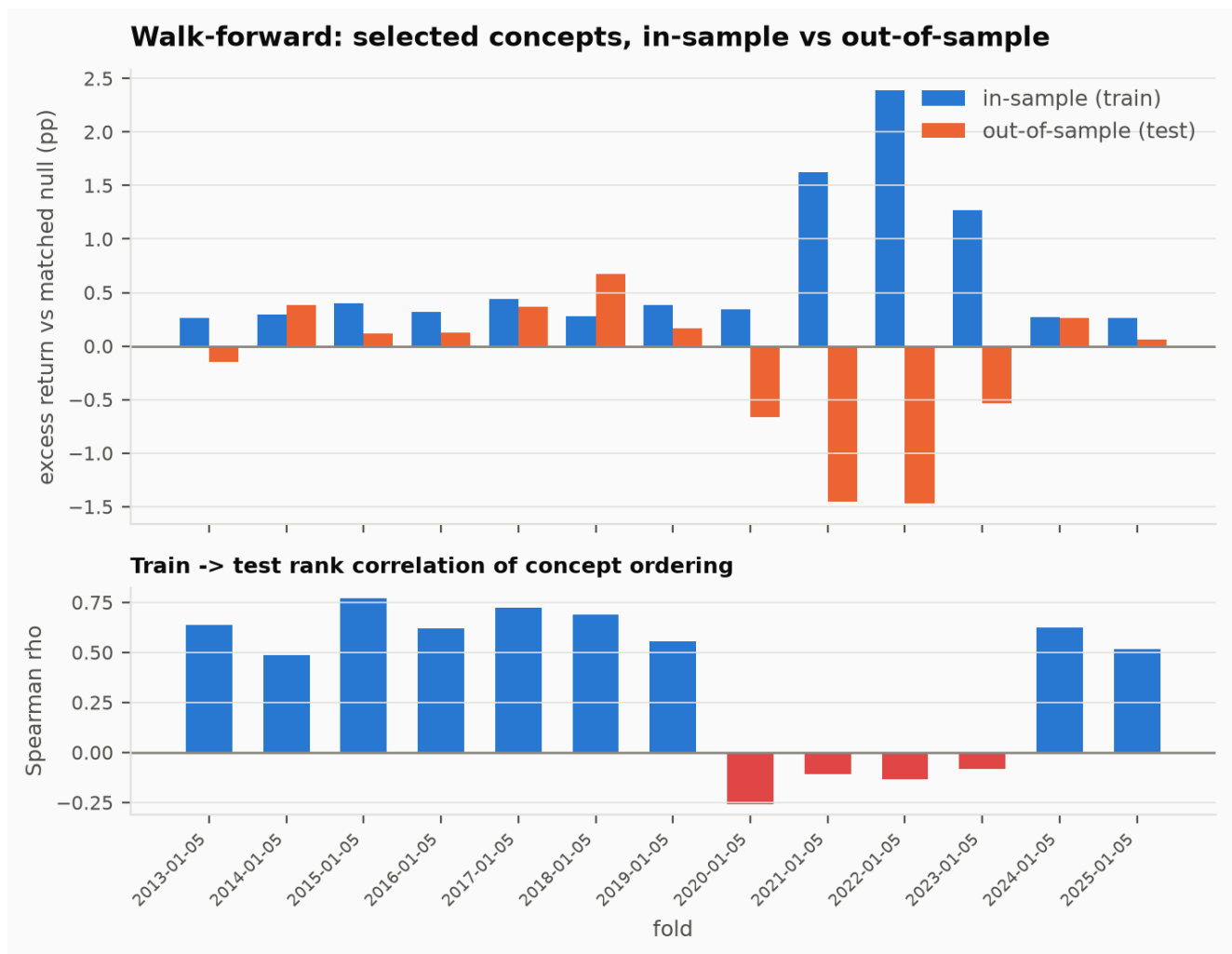


Figure 6 breakeven costs

Cost at which each concept's edge over random entry disappears

